

Amtrak and Chicago Hub Operations

Roy Deitchman and Ellen Jurczak

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Global Climate Change and the Transportation Sector

- As policymakers at state and federal levels address the issue of global climate change, there is consideration to significantly reduce greenhouse gas (GHG) emissions
- Some proposals include an 80% reduction in GHG emissions by 2050.
- Approximately 1/3 of GHG emissions come from the transportation sector in US.
- Energy efficiency is a low cost alternative to developing low-carbon and zero-carbon alternative fuels.

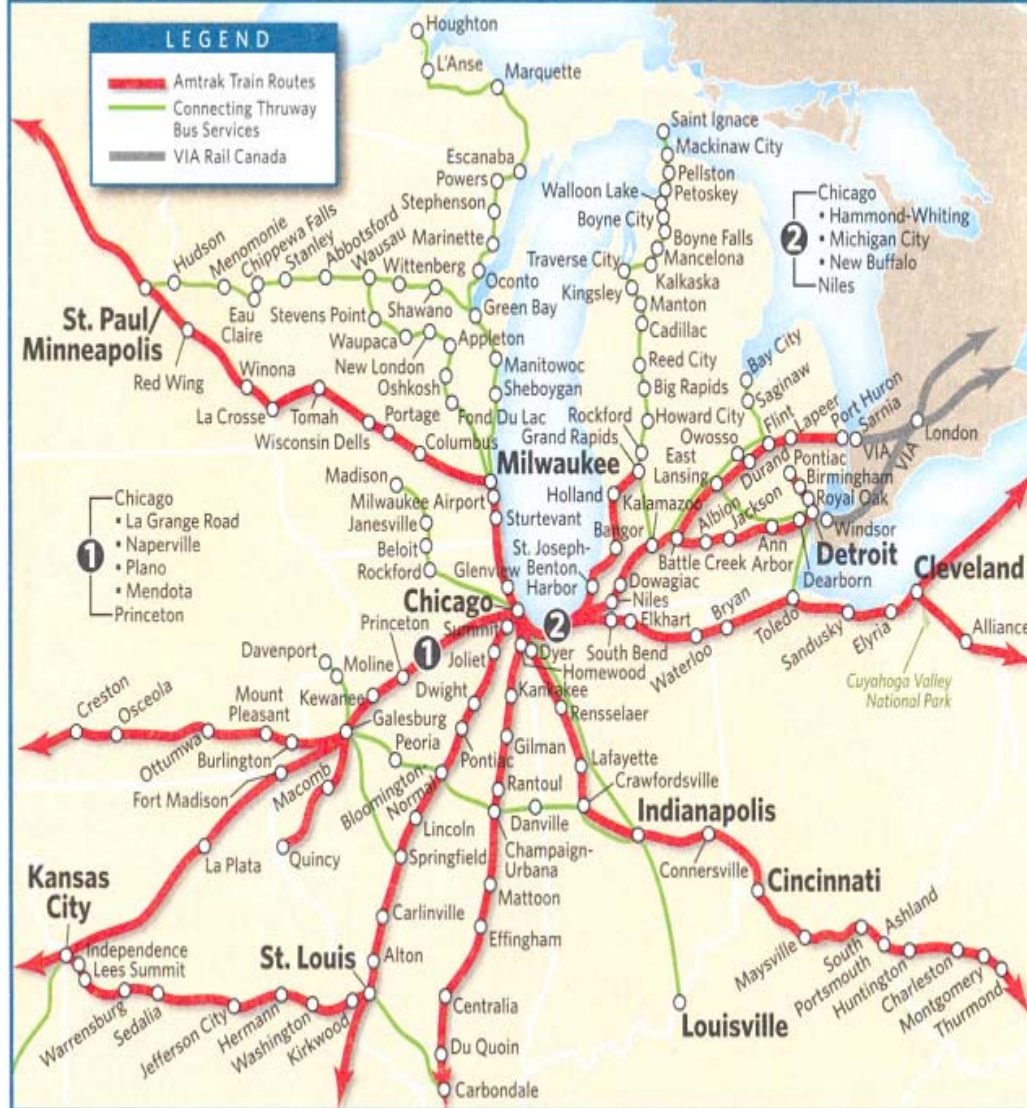


Amtrak's Place In Domestic Transportation

- Amtrak is the nation's principal intercity rail passenger carrier
- Operates 22,000 route-miles of service in 46 states
- Carried over 27 million riders in FY09
- In 2008, Auto-Train service between Virginia and Florida removed 105,439 cars from the 1-95 corridor (train carries cars and passengers)



MIDWEST — TRAIN AND THRUWAY SERVICE



Host Railroads

CHICAGO - ST. LOUIS - KANSAS CITY

Chicago – Joliet
 Canadian National (CN)
 Joliet – Kansas City
 Union Pacific (UP)

ILLINOIS ZEPHYR

Chicago – Quincy
 BNSF Railway Company (BNSF)

HIAWATHA SERVICE

Chicago – Glenview
 (METRA)
 Glenview – Milwaukee
 Canadian Pacific (CP)

WOLVERINE, BLUE WATER AND PERE MARQUETTE SERVICE

Chicago – Porter
 Norfolk Southern (NS)

PERE MARQUETTE SERVICE

Porter – Grand Rapids
 CSX Transportation (CSX)

BLUE WATER AND WOLVERINE SERVICE

Porter – Kalamazoo
 Amtrak
 Kalamazoo – Battle Creek
 Norfolk Southern (NS)

BLUE WATER SERVICE

Battle Creek – Port Huron
 Canadian National (CN)

WOLVERINE SERVICE

Battle Creek – Detroit
 Norfolk Southern (NS)
 Detroit – Pontiac
 Canadian National (CN)

ILLINI

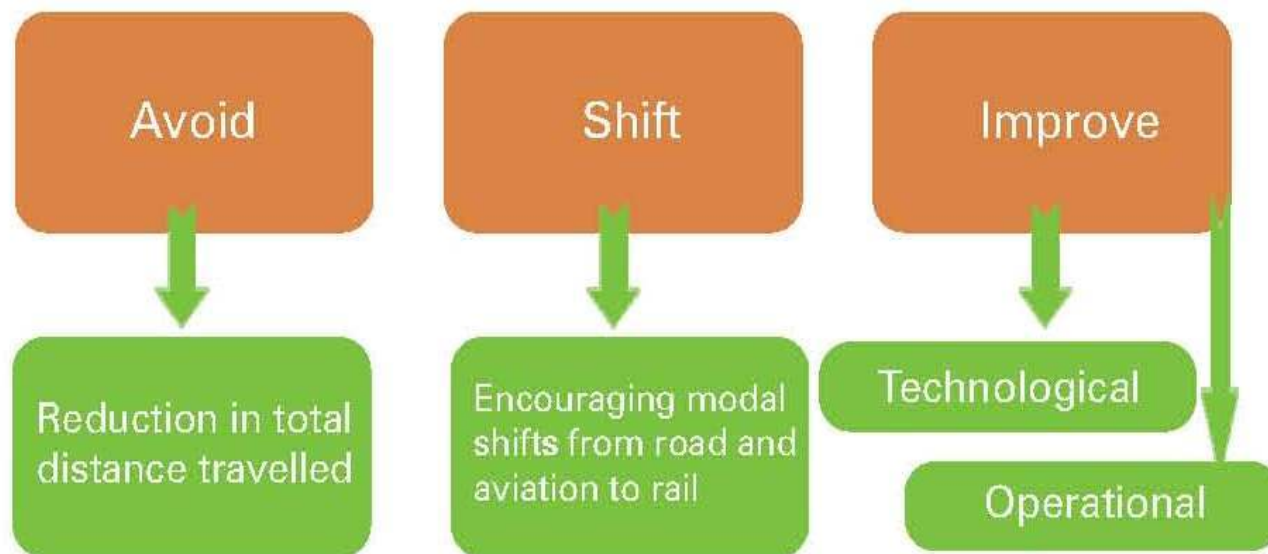
Chicago – Carbondale
 Canadian National (CN)

CITY OF NEW ORLEANS

Chicago – New Orleans



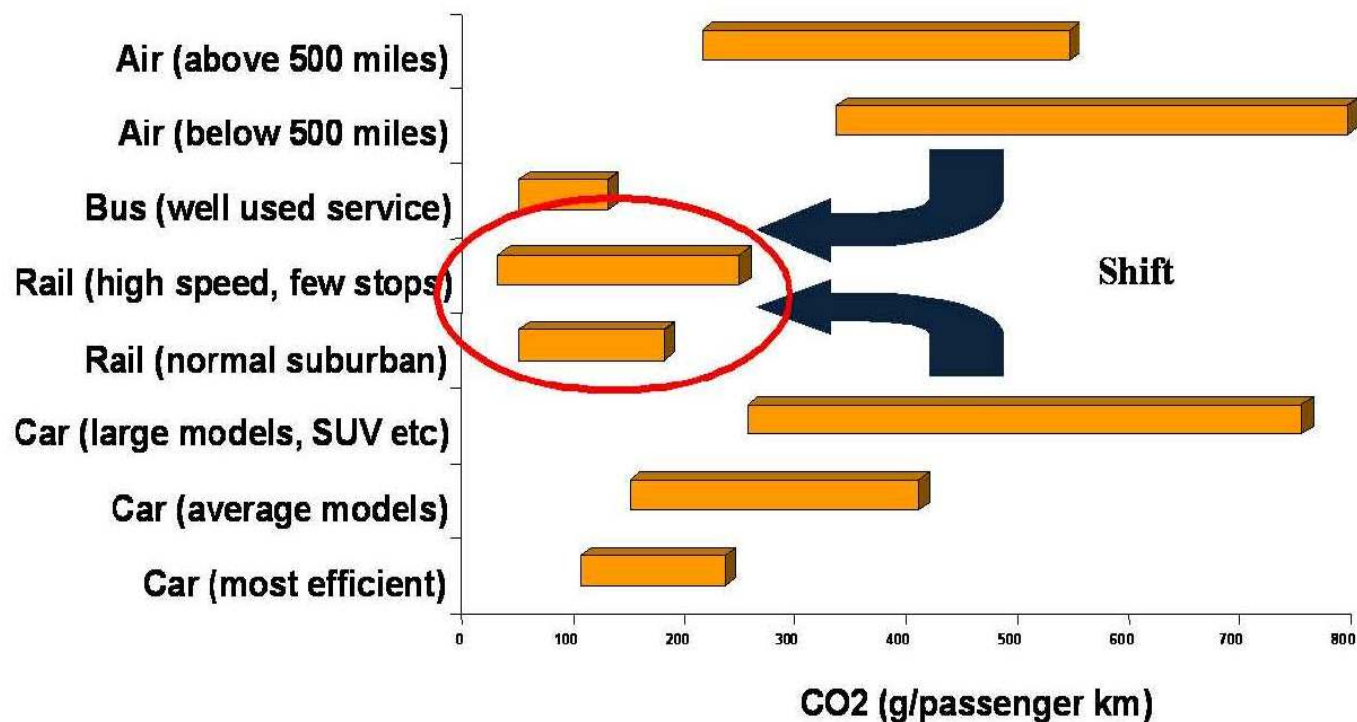
Reducing the Environmental Impact of Transport



Potential rail strategy responses to reduce environmental impacts. *From “Keeping Climate Solutions on Track: The Role of Rail” a global position paper by Kathryn Avery, Holger Dalkmann, Dani Myers, Shaneen Khambata and Margrethe Sagevik.*



Ranges of CO₂ Emissions Across Different Modes of Transport



Chicago Climate Exchange (CCX)

- **Operates the only “cap and trade” system in North America**
- **Members voluntarily agree to reduce their GHG emissions by 6 percent from 2003-2010 from their 1998-2001 baseline**
- **Third party independent verification by the Financial Industry Regulatory Authority (FINRA, formerly NASD)**
- **Membership commitment is voluntary but legally binding**



Amtrak's Voluntary Commitment to Emissions Reduction

- Amtrak is committed to a phased reduction in its diesel emissions levels
- The baseline for the reduction is an average of 1998-2001 emissions levels

Calendar Year	Emissions Commitment in Metric Tons of CO ₂
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Calendar Year	Emissions Commitment in Metric Tons of CO ₂
2004	810,000 (actual: 731,400)
2005	801,700 (actual: 723,100)
2006	793,400 (actual: 668,200)
2007	791,400 (actual: 678,400)
2008	789,300 (actual: 669,600)



GHG Inventory: Overview

- **The Climate Registry's General Reporting Protocol**
- **Must report emissions of six GHGs:**
 - Carbon dioxide (CO₂)
 - Methane (CH₄)
 - Nitrous oxide (N₂O)
 - Sulfur hexafluoride (SF₆)
 - Hydrofluorocarbons (HFCs)
 - Perfluorocarbons (PFCs)
- **Must report both direct and indirect emissions down to the facility level**
- **Obtain third-party verification for each annual report**



Amtrak's Carbon Offset Program

- Program operated by Carbonfund
- Offsets are available to passengers (voluntary) for:
 - 2500 miles (992 pounds of CO₂) of travel at \$4.50
 - 5500 miles (2182 pounds of CO₂) of travel at \$10.00
- Program started in April 2008
- Through June 2009, over 8,000,000 miles of offsets have been purchased by Amtrak passengers



Diesel Fuel Reduction Methods

- Anti-idling practices
- Automatic Start/Stop installation
- Aerodynamic improvements of rolling stock
- Consist/Locomotive management
- Locomotive upgrades and improved maintenance
- Locomotive Engineer Training



Amtrak is a registered service mark of the National Railroad Passenger Corporation.



Electric Locomotives

- **Higher BTU Use**
 - Some power loss in transmission lines
- **Higher Speeds**
- **Regenerative Braking**
- **Carbon Emissions Dependent on Type of Power Purchased**
 - Electricity generated by coal yields higher carbon emissions
 - Nuclear, Wind and Hydro-electric power have limited carbon emissions
 - Emissions of other contaminants likely lower due to “stack controls” on power plants





Diesel Locomotives

- **Engine drives an electric generator that runs electric motor**
- **High Horsepower, over 4000 hp**
 - 12 to 16 cylinder engines
- **Manufactured by General Electric and EMD**
- **Installing 480 volt plug-ins to provide Head End Power (HEP) for service at layover stations**





Zero Emissions Rail Travel

